Math 54 — Week 4

Solutions

P1.

Idea. Use the row-column rule; the transposes check the rule $(AB)^T = B^T A^T$.

1. By the row-column rule,

$$AB = \begin{bmatrix} 0-4 & 2+2 \\ 0+12 & 1-6 \end{bmatrix} = \begin{bmatrix} -4 & 4 \\ 12 & -5 \end{bmatrix}, \qquad BA = \begin{bmatrix} 0+1 & 0+3 \\ 8-2 & -4-6 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 6 & -10 \end{bmatrix}.$$

   The two products differ.

2. Transposing AB, and multiplying the transposes in the reverse order:

$$(AB)^T = \begin{bmatrix} -4 & 12 \\ 4 & -5 \end{bmatrix}, \qquad B^T A^T = \begin{bmatrix} 0 & 4 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 0-4 & 0+12 \\ 2+2 & 1-6 \end{bmatrix} = \begin{bmatrix} -4 & 12 \\ 4 & -5 \end{bmatrix}.$$

   They agree, as Theorem 3 says.

P2.

Idea. X is a horizontal shear by -2; doing it k times shears by $-2k$. Prove the formula by induction.

1. By the row-column rule,

$$X^2 = \begin{bmatrix} 1 & -4 \\ 0 & 1 \end{bmatrix}, \qquad X^3 = X^2 X = \begin{bmatrix} 1 & -6 \\ 0 & 1 \end{bmatrix}.$$

2. Guess: for every $k \geq 0$,

$$X^k = \begin{bmatrix} 1 & -2k \\ 0 & 1 \end{bmatrix}.$$

3. For $k = 0$ this is $I = X^0$. If it holds for k, then

$$X^{k+1} = X^k X = \begin{bmatrix} 1 & -2k \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -2-2k \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -2(k+1) \\ 0 & 1 \end{bmatrix},$$

   so it holds for $k+1$. By induction it holds for all k.

P3.

Idea. One inverse serves every right-hand side; for $\mathbf{b} = \mathbf{e}_2$ the solution is column 2 of A^{-1}.

1. The coefficient matrix has determinant $2 \cdot 3 - 5 \cdot 1 = 1$, so by Theorem 4

$$A = \begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix}, \qquad A^{-1} = \begin{bmatrix} 3 & -5 \\ -1 & 2 \end{bmatrix}.$$

2. First system:

$$\mathbf{x} = A^{-1} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 3+5 \\ -1-2 \end{bmatrix} = \begin{bmatrix} 8 \\ -3 \end{bmatrix}.$$

   Check: $16 - 15 = 1$ and $8 - 9 = -1$.

3. Second system: $A^{-1}\mathbf{e}_2$ is column 2 of A^{-1}, so $\mathbf{x} = (-5, 2)$. Check: $-10 + 10 = 0$ and $-5 + 6 = 1$.

P4.

Idea. Row reduce $[\,A \;\; I\,]$; row 2 has leading entry 1, so interchange first.

1. Forward phase:

$$\left[\begin{array}{ccc|ccc} 2 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 3 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 0 & 1 & 0 \\ 2 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 3 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_2 - 2R_1} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & -1 & -2 & 1 & -2 & 0 \\ 0 & 1 & 3 & 0 & 0 & 1 \end{array}\right]$$

$$\xrightarrow{R_3 + R_2} \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & -1 & -2 & 1 & -2 & 0 \\ 0 & 0 & 1 & 1 & -2 & 1 \end{array}\right]$$

2. Backward phase:

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & -1 & -2 & 1 & -2 & 0 \\ 0 & 0 & 1 & 1 & -2 & 1 \end{array}\right] \xrightarrow[R_1 - R_3]{R_2 + 2R_3} \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & -1 & 3 & -1 \\ 0 & -1 & 0 & 3 & -6 & 2 \\ 0 & 0 & 1 & 1 & -2 & 1 \end{array}\right]$$

$$\xrightarrow{-R_2} \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & -1 & 3 & -1 \\ 0 & 1 & 0 & -3 & 6 & -2 \\ 0 & 0 & 1 & 1 & -2 & 1 \end{array}\right] \xrightarrow{R_1 - R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 2 & -3 & 1 \\ 0 & 1 & 0 & -3 & 6 & -2 \\ 0 & 0 & 1 & 1 & -2 & 1 \end{array}\right]$$

3. So

$$A^{-1} = \begin{bmatrix} 2 & -3 & 1 \\ -3 & 6 & -2 \\ 1 & -2 & 1 \end{bmatrix}.$$

4. Check AA^{-1} column by column with the row-column rule:

$$AA^{-1} = \begin{bmatrix} 4-3+0 & -6+6+0 & 2-2+0 \\ 2-3+1 & -3+6-2 & 1-2+1 \\ 0-3+3 & 0+6-6 & 0-2+3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Since A is square, $AA^{-1} = I$ is enough (first consequence of the IMT).

P5.

Idea. Look for a dependence among the rows or columns before computing anything.

1. Row 2 is twice row 1, so $R_2 - 2R_1$ produces a zero row, and an echelon form has at most 2 pivots.
2. Fewer than 3 pivots means statement (c) of the invertible matrix theorem fails, so the matrix is not invertible.

P6.

Idea. Column 2 has a single nonzero entry; after that, use the column of the minor with the most zeros.

1. Expand down column 2: only $a_{32} = 7$ is nonzero, with sign $(-1)^{3+2} = -1$. So

$$\det A = -7 \det M, \qquad M = \begin{bmatrix} 3 & 4 & 1 \\ 2 & -1 & 0 \\ 1 & 2 & 0 \end{bmatrix}.$$

2. Expand $\det M$ down its column 3: only the top entry 1 is nonzero, with sign $(-1)^{1+3} = +1$:

$$\det M = 1 \cdot \det \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix} = 4 + 1 = 5.$$

3. So $\det A = -7 \cdot 5 = -35$.

P7.

Idea. Each part is one implication of the invertible matrix theorem or of Theorem 6.

1. (a) True. A nontrivial solution needs a free variable, that is, a column without a pivot, so there are fewer than n pivots.
2. (b) True for square matrices: (h)⇒(e) in the invertible matrix theorem.
3. (c) True. By (k), A is invertible; then $B = A^{-1}(AB) = A^{-1}$, which is invertible by Theorem 6(a).
4. (d) True. By Theorem 6(b) with $B = A$, $A^2 = AA$ is invertible with inverse $A^{-1}A^{-1}$.

P8.

Idea. Multiply out and use $A^3 = 0$; for square matrices one product equal to I is enough.

1. By the distributive laws,

$$(I - A)(I + A + A^2) = I + A + A^2 - A - A^2 - A^3 = I - A^3 = I.$$

2. $I - A$ and $I + A + A^2$ are square, so by the first consequence of the invertible matrix theorem, $I - A$ is invertible and $(I - A)^{-1} = I + A + A^2$.